

Projected Health Impact of Achieving WHO Targets for Cardiovascular Risk-Factor Control

Deterministic sensitivity analysis (one-way scenario analyses)

Work in Progress

WHO · University of Washington

September 2026

Study Overview

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Study Overview

Aim 1 Deterministic Sensitivity Analysis

Question

How robust are the projected deaths averted by the WHO cardiovascular targets to **achievement, timing, statin adherence** and **CVD mortality-trend** assumptions?

Design (Briggs et al., ISPOR-SMDM 2012)

- One-way deterministic scenarios around the **active base case**
- Every setting re-runs the country-level model (186 countries, 6 scenarios, 2017–2050)
- Chosen scenario values, **not** probability distributions: ranges are **not** confidence intervals

Reference = active base case

BP: 150M additional controlled by 2030 · statins 50% by 2030 · statin adherence IR/CF **0.575 / 0.664** (not 1.0) · CVD CF trend at 80% · background trend on

DSA Settings

Setting	Family	What changes
Reference	Reference	—
50% of increment	Target achievement	50% of increment by 2030
75% of increment	Target achievement	75% of increment by 2030
Endpoint 2035	Implementation timing	full endpoint in 2035
Endpoint 2040	Implementation timing	full endpoint in 2040
Statin adherence low	Statin adherence	statin adherence 0.50 / 0.65
Statin adherence moderate	Statin adherence	statin adherence 0.70 / 0.80
Statin adherence by income	Statin adherence by income group	statin adherence by income group (illustrative)
Perfect adherence (extra ref.)	Extra reference (not base case)	statin adherence 1.00 / 1.00
Full CVD CF trend	CVD mortality trend	trend_full_cvd
No CVD CF trend	CVD mortality trend	trend_no_cvd
IHME Foresight CVD trend	CVD mortality trend	trend_ihme_cvd

Income-group adherence (L 0.50/0.65, LM 0.60/0.70, UM 0.70/0.80, H 0.80/0.90) is an **illustrative assumption**.
 Perfect adherence (1.00/1.00) is an extra reference, never the anchor.

Scenarios

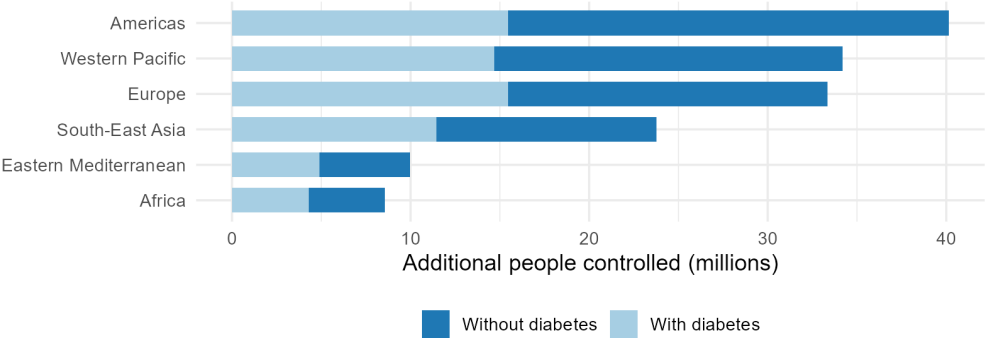
Four WHO Priority Targets (Reference)

#	WHO Target	Modelled Scenario
1	150M more with HTN controlled by 2030	HTN Control (Combined): 150.0M = 83.8M without + 66.2M with diabetes
2	$\geq 50\%$ eligible adults on lipid-lowering therapy by 2030	Improved Statin Uptake: 50% coverage by 2030
3	80% BP control among diagnosed diabetics by 2030	HTN Control (Diabetes): 0.80 ceiling (full target only)
4	80% availability of validated BP devices	Embedded in HTN scenarios

The two BP subgroups combine additively; the BP composite combines multiplicatively with statins.

The 150 Million Split by WHO Region and Diabetes Status

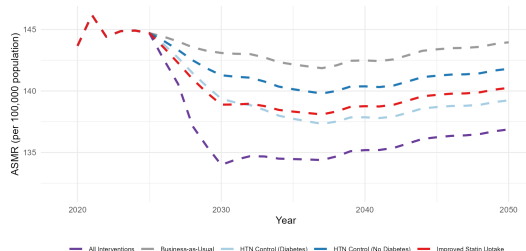
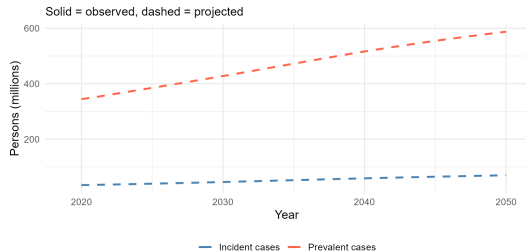
Fixed 2025 population | total 150.0 M



Region	HTN pop 2025 (M)	Cov. 2025	Cov. 2030	Added (M)
Africa	182.0	11.3%	16.0%	8.5
Americas	273.9	36.7%	51.3%	40.1
Eastern Mediterranean	157.5	15.2%	21.6%	10.0
Europe	300.3	26.8%	37.9%	33.3
South-East Asia	425.8	13.4%	19.0%	23.8

Reference Results

Baseline Disease Trends and ASMR (Business-as-Usual)



Reference trend regime; solid = calibration, dashed = projection.

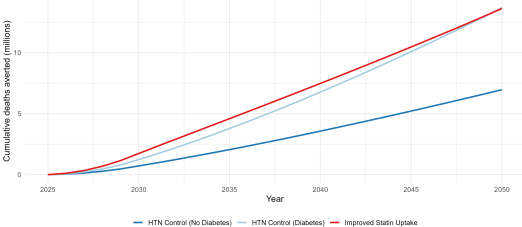
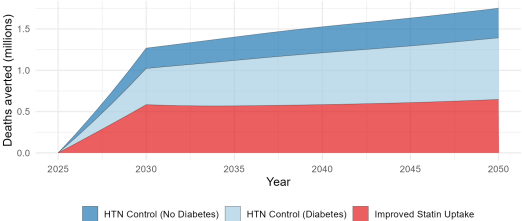
Summary: Cumulative Deaths Averted, 2025–2050 (Reference)

Scenario	Cumulative (M)	Avg./year (M)	% of BAU deaths
HTN Control (No Diabetes)	7.0 M	0.3 M	1%
HTN Control (Diabetes)	13.7 M	0.5 M	2%
Improved Statin Uptake	13.6 M	0.5 M	2%
All Interventions	26.8 M	1.1 M	3.9%

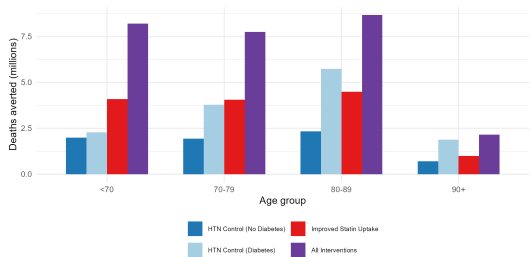
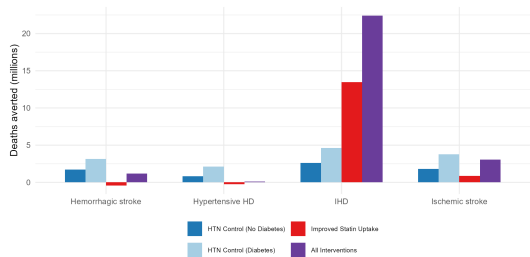
Headline (reference)

All Interventions: 26.8 million deaths averted, 2025–2050 ($\approx$ 1.1 M/year).

Annual and Cumulative Deaths Averted (Reference)



Deaths Averted by Cause and Age (Reference)



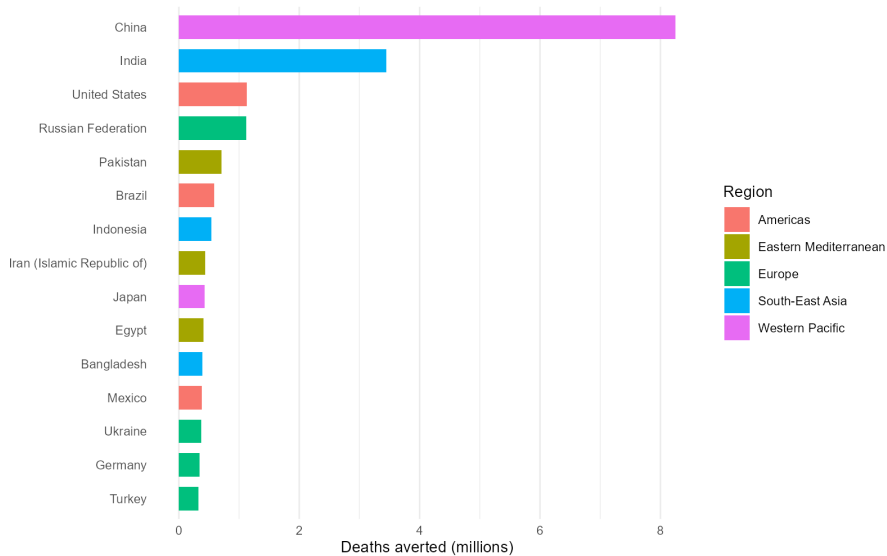
Burden of Disease Averted and Regional Summary (Reference)

Intervention	Deaths (M)	YLLs (M)	YLDs (M)	DALYs (M)
HTN Control (No Diabetes)	7.0	89.7	2.0	91.7
HTN Control (Diabetes)	13.7	135.1	4.5	139.6
Improved Statin Uptake	13.6	170.6	-1.7	168.9
All Interventions	26.8	348.5	-11.4	337.1

Region	BAU (M)	Averted (M)	% averted
Western Pacific	225.9	9.8	4.3%
South-East Asia	175.8	5.0	2.9%
Europe	117.2	4.5	3.9%
Americas	62.6	3.3	5.2%
Eastern Mediterranean	64.0	2.7	4.2%
Africa	48.7	1.5	3.0%

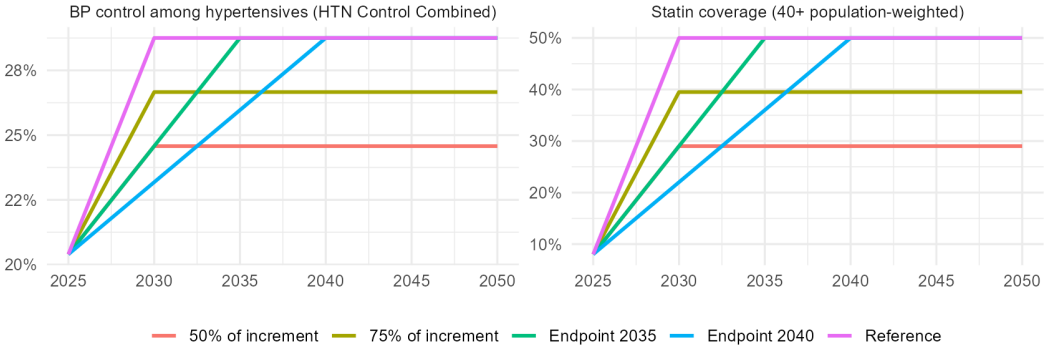
DALYs 2026–2050 (YLL: WPP 2024 life expectancy; YLD: GBD 2023 disability weights); regions 2025–2050, All Interventions.

Top Countries by Projected Impact (Reference)



Deterministic Sensitivity Analysis

Coverage Trajectories: Partial Achievement and Delayed Implementation



Adherence and trend settings share the reference trajectories.

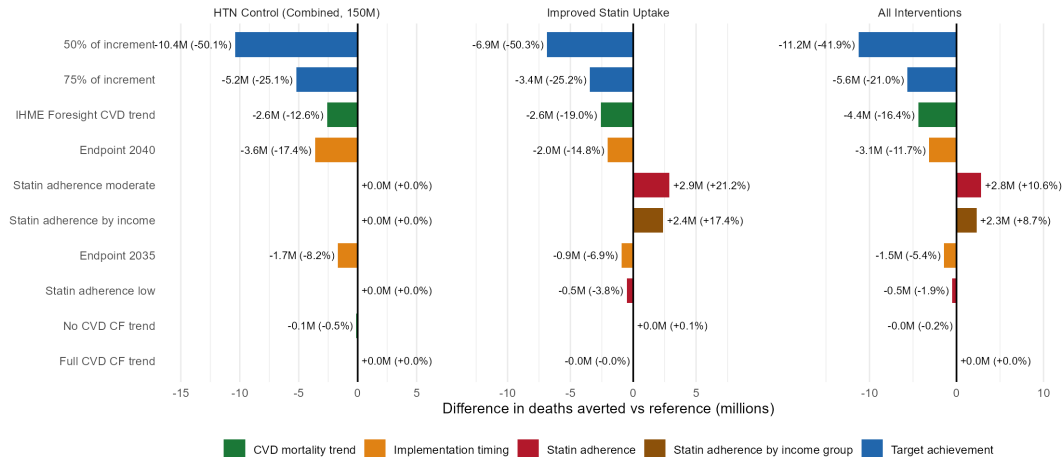
Adherence and trend settings share the reference trajectories. Additional controlled people use the fixed 2025 population of script 04.

Achieved BP Control and Statin Inputs

DSA setting	BP endpoint	Added 2030 (M)	2035 (M)	2040 (M)	At endpoint (M)	Share of full target	Diabetes control at endpoint	Statin endpoint	Mean statin baseline / endpoint
Reference	2030	150.00	150.00	150.00	150.00	100.0%	45.3%	2030	7.6% / 50.0%
50% of increment	2030	75.00	75.00	75.00	75.00	50.0%	38.8%	2030	7.6% / 28.8%
75% of increment	2030	112.50	112.50	112.50	112.50	75.0%	42.1%	2030	7.6% / 39.4%
Endpoint 2035	2035	75.00	150.00	150.00	150.00	100.0%	45.3%	2035	7.6% / 50.0%
Endpoint 2040	2040	50.00	100.00	150.00	150.00	100.0%	45.3%	2040	7.6% / 50.0%

Partial settings do not reach 150 million: they reach the stated share of the full increment. The diabetic 0.80 figure is the full-target ceiling; the achieved population-weighted diabetes control is shown.

One-Way Results: Change in Deaths Averted vs Reference



Cumulative 2026–2050; deterministic scenario differences, **not** intervals. Trend settings are compared with their own BAU.

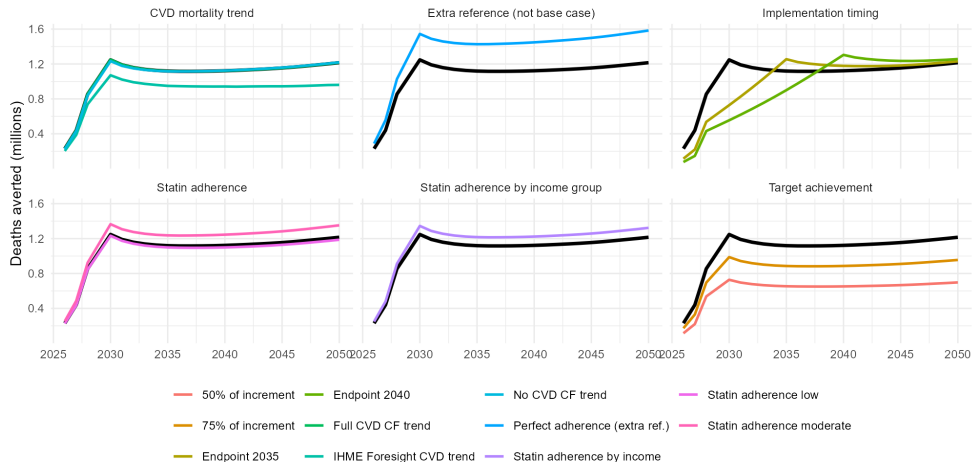
One-Way Results: All Interventions

Setting	Family	Averted (M)	Δ vs ref. (M)	Δ (%)	% of BAU
Reference	Reference	26.79	—	—	4.0%
50% of increment	Target achievement	15.56	-11.23	-41.9%	2.3%
75% of increment	Target achievement	21.16	-5.63	-21.0%	3.1%
Endpoint 2035	Implementation timing	25.33	-1.46	-5.4%	3.7%
Endpoint 2040	Implementation timing	23.65	-3.14	-11.7%	3.5%
Statin adherence low	Statin adherence	26.28	-0.51	-1.9%	3.9%
Statin adherence moderate	Statin adherence	29.63	+2.84	+10.6%	4.4%
Statin adherence by income	Statin adherence by income group	29.11	+2.32	+8.7%	4.3%
Perfect adherence (extra ref.)	Extra reference (not base case)	34.25	+7.46	+27.9%	5.1%
Full CVD CF trend	CVD mortality trend	26.80	+0.01	+0.0%	4.0%
No CVD CF trend	CVD mortality trend	26.74	-0.05	-0.2%	4.0%
IHME Foresight CVD trend	CVD mortality trend	22.40	-4.39	-16.4%	4.1%

Range across one-way settings (All Interventions, 2026–2050)

15.6 M (50% of increment) to 29.6 M (Statin adherence moderate) vs 26.8 M in the reference — a deterministic spread, not a confidence interval.

Annual Deaths Averted by DSA Setting



Endpoints: 2030, 2035, 2040, 2050 (All Interventions)

DSA setting	Annual 2030	Annual 2035	Annual 2040	Annual 2050	Cumulative 2026–2030	Cumulative 2026–2035	Cumulative 2026–2040	Cumulative 2026–2050
Reference	1.248	1.119	1.122	1.215	3.822	9.554	15.144	26.788
50% of increment	0.728	0.654	0.652	0.698	2.232	5.585	8.840	15.560
75% of increment	0.988	0.886	0.886	0.955	3.028	7.567	11.986	21.158
Endpoint 2035	0.728	1.255	1.178	1.231	2.232	7.416	13.394	25.330
Endpoint 2040	0.555	0.900	1.304	1.256	1.702	5.485	11.180	23.652
Statin adherence low	1.233	1.099	1.099	1.186	3.780	9.421	14.898	26.279
Statin adherence moderate	1.365	1.237	1.245	1.352	4.173	10.491	16.684	29.626
Statin adherence by income	1.345	1.217	1.223	1.323	4.114	10.332	16.420	29.106
Perfect adherence (extra ref.)	1.545	1.427	1.448	1.584	4.703	11.948	19.131	34.251
Full CVD CF trend	1.251	1.119	1.122	1.215	3.835	9.573	15.161	26.798
No CVD CF trend	1.233	1.116	1.125	1.219	3.765	9.462	15.055	26.742
IHME Foresight CVD trend	1.071	0.949	0.942	0.961	3.306	8.200	12.916	22.399

Deaths averted (millions): annual in the endpoint year and cumulative from 2026.

Statin Adherence by World Bank Income Group (Illustrative)

Intervention	Income group	Reference	Income-group adherence	Δ (M)	Δ (%)
All Interventions	Low income	1.158	1.137	-0.022	-1.9%
All Interventions	Lower-middle income	7.287	7.521	+0.234	+3.2%
All Interventions	Upper-middle income	13.079	14.450	+1.372	+10.5%
All Interventions	High income	5.264	5.998	+0.734	+13.9%
All Interventions	Global	26.788	29.106	+2.318	+8.7%
Improved Statin Uptake	Low income	0.570	0.548	-0.022	-3.9%
Improved Statin Uptake	Lower-middle income	4.420	4.655	+0.236	+5.3%
Improved Statin Uptake	Upper-middle income	6.655	8.059	+1.403	+21.1%
Improved Statin Uptake	High income	1.979	2.727	+0.747	+37.8%
Improved Statin Uptake	Global	13.625	15.989	+2.364	+17.4%

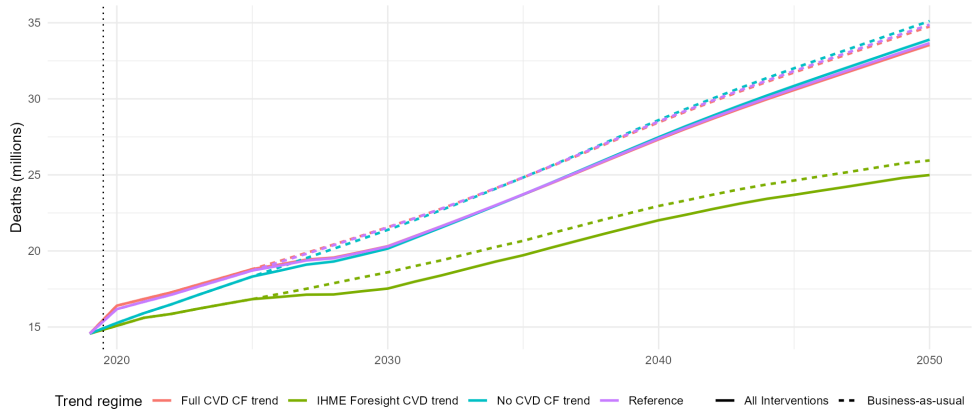
Adherence changes statin incidence (IR) and case-fatality (CF) effects only; BP-only scenarios are unchanged (verified for every country).

Alternative CVD Mortality Trends (Structural)

Mortality trend	Intervention	Projected deaths (M)	BAU deaths (M)	Deaths averted (M)	Averted per year (M)	Averted (% of BAU)	Δ averted vs reference (M)	Δ (%)
Reference	HTN Control (Combined, 150M)	654.86	675.56	20.70	0.828	3.1%	—	—
Reference	Improved Statin Uptake	661.93	675.56	13.62	0.545	2.0%	—	—
Reference	All Interventions	648.77	675.56	26.79	1.072	4.0%	—	—
Full CVD CF trend	HTN Control (Combined, 150M)	653.93	674.63	20.70	0.828	3.1%	+0.01	+0.0%
Full CVD CF trend	Improved Statin Uptake	661.01	674.63	13.62	0.545	2.0%	-0.01	-0.0%
Full CVD CF trend	All Interventions	647.83	674.63	26.80	1.072	4.0%	+0.01	+0.0%
No CVD CF trend	HTN Control (Combined, 150M)	655.56	676.14	20.58	0.823	3.0%	-0.11	-0.5%
No CVD CF trend	Improved Statin Uptake	662.50	676.14	13.64	0.545	2.0%	+0.01	+0.1%
No CVD CF trend	All Interventions	649.40	676.14	26.74	1.070	4.0%	-0.05	-0.2%
IHME Foresight CVD trend	HTN Control (Combined, 150M)	528.17	546.26	18.09	0.724	3.3%	-2.60	-12.6%
IHME Foresight CVD trend	Improved Statin Uptake	535.22	546.26	11.04	0.442	2.0%	-2.58	-19.0%
IHME Foresight CVD trend	All Interventions	523.86	546.26	22.40	0.896	4.1%	-4.39	-16.4%

Background trend on in every regime; each regime compared with its own BAU. Layout after Table S8 of the related RTSL draft.

CVD Deaths by Trend Regime: BAU and All Interventions



Burden of Disease Averted by DSA Setting

DSA setting	Deaths (M)	YLLs (M)	YLDs (M)	DALYs (M)	Δ DALYs vs ref. (M)	Δ (%)
Reference	26.79	348.50	-11.39	337.11	—	—
50% of increment	15.56	208.19	-6.66	201.52	-135.59	-40.2%
75% of increment	21.16	278.23	-9.02	269.21	-67.90	-20.1%
Endpoint 2035	25.33	327.30	-10.30	317.00	-20.11	-6.0%
Endpoint 2040	23.65	303.81	-9.16	294.64	-42.47	-12.6%
Statin adherence low	26.28	342.52	-11.76	330.76	-6.35	-1.9%
Statin adherence moderate	29.63	383.54	-11.68	371.86	+34.75	+10.3%
Statin adherence by income	29.11	375.55	-11.62	363.93	+26.82	+8.0%
Perfect adherence (extra ref.)	34.25	439.92	-11.35	428.57	+91.46	+27.1%
Full CVD CF trend	26.80	348.66	-11.43	337.23	+0.12	+0.0%
No CVD CF trend	26.74	347.42	-11.25	336.17	-0.94	-0.3%
IHME Foresight CVD trend	22.40	275.98	-9.08	266.90	-70.21	-20.8%

All Interventions, 2026–2050, each setting against its own BAU. Negative YLDs averted: people kept alive live longer with CVD (as in the ordinary report).

Economic Value by DSA Setting

DSA setting	VSL (USD T)	VSL % GNI	VSly (USD T)	VSly % GNI	Δ VSly vs ref.
Reference	80.51	2.0%	32.45	0.8%	—
50% of increment	45.74	1.1%	18.89	0.5%	-41.8%
75% of increment	63.08	1.5%	25.66	0.6%	-20.9%
Endpoint 2035	75.17	1.8%	30.12	0.7%	-7.2%
Endpoint 2040	69.27	1.7%	27.63	0.7%	-14.8%
Statin adherence low	79.20	1.9%	31.95	0.8%	-1.5%
Statin adherence moderate	88.07	2.1%	35.41	0.9%	+9.2%
Statin adherence by income	89.24	2.2%	35.80	0.9%	+10.3%
Perfect adherence (extra ref.)	100.35	2.4%	40.19	1.0%	+23.9%
Full CVD CF trend	80.53	2.0%	32.46	0.8%	+0.0%
No CVD CF trend	80.36	2.0%	32.35	0.8%	-0.3%
IHME Foresight CVD trend	66.41	1.6%	25.58	0.6%	-21.2%

All Interventions, World, cumulative 2026–2050; VSL/VSLY primary elasticity (0.8/1.2), discounted at 3% to 2026; recomputed from each setting's own deaths averted.

Economic Value of Avoidable Mortality (Reference)

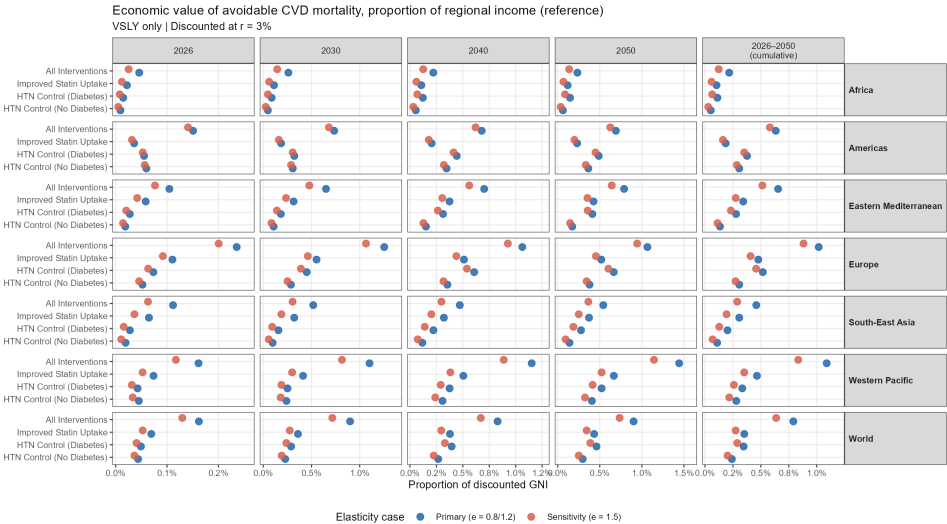
Economic Value by WHO Region (Primary, Cumulative 2026–2050)

World: All Interventions (reference, primary, 3%)

VSLY = 0.8% of annual income PPP (cumulative **\$32.4 T**).

Region	VSL (USD T)	VSL % GNI	VSLY (USD T)	VSLY % GNI
Africa	1.02	0.5%	0.46	0.2%
Americas	15.63	1.7%	5.73	0.6%
Eastern Mediterranean	4.07	1.4%	1.96	0.7%
Europe	22.87	2.5%	9.19	1.0%
South-East Asia	6.80	1.0%	3.16	0.5%
Western Pacific	30.12	2.7%	11.95	1.1%
World	80.51	2.0%	32.45	0.8%

Value of Avoidable Mortality as a Proportion of Income (Reference)



Discussion

Interpretation

What the one-way settings show

- Target achievement (50%/75% of the increment) scales benefits roughly in proportion to the achieved increment
- Delaying the full endpoint to 2035/2040 loses early benefits; effects converge once the endpoint is reached
- Statin adherence moves statin-driven benefits only; BP benefits are unchanged
- The CVD case-fatality trend regime changes both BAU and intervention deaths; compare within regimes

Interpretation rule

Deterministic scenario spreads describe the consequences of named assumptions; they are not probability statements or confidence intervals.

Limitations of the DSA

- One-way settings only; no multi-way stress tests or probability weights
- Income-group statin adherence values are illustrative (no sourced income-specific estimate in the repository)
- No separate BP-medication adherence: BP targets are achieved control, which already embodies adherence
- The cause-specific background-mortality trend input has no post-2019 values, so only the all-cause envelope (BG.mx.all) trends, in every regime alike
- The reference CVD trend series starts above the 2019 level for IHD and ischemic stroke; the IHME Foresight trend is global, cause-level and age-standardised
- Active statin CF adherence 0.664 vs 64.4% reported for secondary prevention by Basios et al. 2025 (kept to reproduce the pipeline)

Validation

Result	n
pass	113
informational	15

Checks by code/10_deterministic_dsa.R (run full): DSA reference bit-identical to the live 06 functions and to the ordinary 06 output; 05 reproduced for every trend regime; trajectories, targets, adherence and trend formulas recomputed independently.

Appendix

Notation: Core Model and HTN Control

Symbol	Description
W, K, D	Well, Sick, Dead populations
IR, CF	Baseline incidence & case-fatality rates
$\varepsilon_{IR,j}, \varepsilon_{CF,j}$	Multiplicative effect ratios for intervention j
$BG.mx, BG.mx.all$	Cause-specific & all-cause background mortality
$c_{g,0}, c_{g,t}, c_{g,target}$	Baseline, time- t , target control, subgroup $g \in \{N, D\}$
f	DSA fraction of the increment: $c_{target}^{DSA} = c_0 + f(c_{target} - c_0)$
T	Endpoint year: $c_t = c_0 + (c_{target} - c_0) \min[\max(0, (t - 2025)/(T - 2025)), 1]$
$e_{g,t}$	$E_g(c_{g,t} - c_{g,0}) / (1 - E_g c_{g,0})$
w_D	Diabetes share among hypertensives; $\varepsilon_{IR,BP} = 1 - (1 - w_D)e_{N,t} - w_De_{D,t}$

Statin Therapy and Adherence

Incremental coverage above baseline $U_{0,\ell}$ (DSA endpoint T):

$$\Delta U(t, \ell) = \Delta U_{\text{target}, \ell} \min \left[\frac{t - 2026 + 1}{T - 2026 + 1}, 1 \right]$$

$$\Delta U_{\text{target}, \ell} = \max(U_{\text{target}}^{DSA} - U_{0, \ell}, 0),$$

$$U_{\text{target}}^{DSA} = U_0 + f(0.50 - U_0)$$

Adherence a_{IR} , a_{CF} enter both numerator and baseline denominator:

$$E^{IR} = \frac{AF \delta^{IR} \Delta U a_{IR}}{1 - \delta^{IR} U_0 a_{IR}}, \quad E^{CF} = \frac{\delta^{CF} \Delta U a_{CF}}{1 - \delta^{CF} U_0 a_{CF}}$$

Reference $a_{IR} = 0.575$, $a_{CF} = 0.664$.

References

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